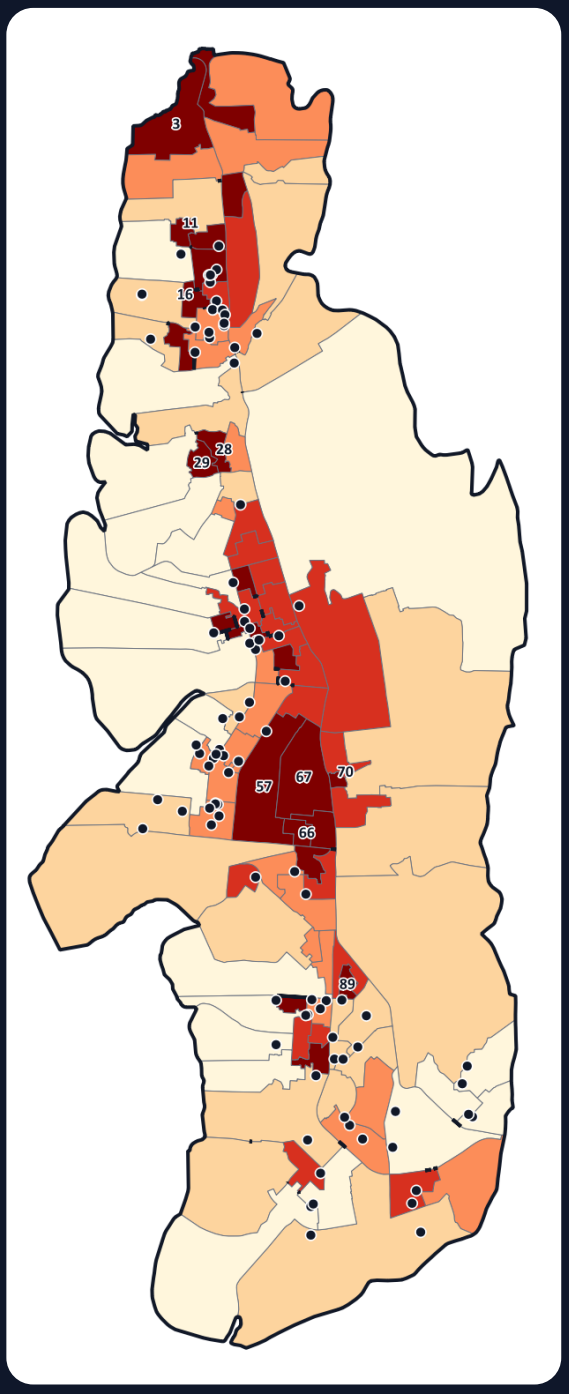
GIS · CA 3 · UNDERSTANDING GUIDE

Urban Climate Stress Mapping of Navi Mumbai

Heat, green cover and flood risk, ward by ward

A plain-language guide with every number

Heat | Green cover | Flood



Combined stress by ward (QGIS map, April 2026)

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Presentation: 30 September 2026

Numbers come from real satellite data: Landsat 8/9, Sentinel-2, ESA WorldCover and SRTM, processed on 26 Sep 2026. QGIS project built on 27 Sep 2026.

The project in 60 seconds

Three stresses, one score, one map

Heat

Which wards are the hottest?

Green cover

Which wards have the least plants?

Flood

Which wards collect water more easily?

UCSI = Urban Climate Stress Index

One number for each of the 111 wards, from 1 (low stress) to 5 (high stress).
It is the average of the heat score, the green-cover-deficit score and the flood score.

A map of Navi Mumbai and 10 queries

Shows where heat, low green cover and flooding overlap.

In one sentence

We use free satellite data to give every ward of Navi Mumbai a heat score, a green-cover score and a flood score, add them into one index, and map where the stresses overlap.

What it is not

It is not a forecast of floods or heat waves. It compares wards with each other. It is a planning aid.

Numbers at a glance

Real results from the QGIS project (April 2026 satellite data)

111

wards analysed, 122.9 km² in total

Sum of the ward polygons in your GeoPackage

13.9 °C

gap between the hottest ward (ward 70: 47.6 °C) and the coolest (ward 111: 33.7 °C)

Ward averages, median of 3 Landsat dates

23%

of the city area is 45 °C or hotter

Area-weighted, inside the ward polygons

33% | 26%

of the area is built-up | tree cover

ESA WorldCover 2021

+0.76

correlation of built-up land with temperature (more concrete, hotter)

Across 111 wards

−0.75

correlation of vegetation (NDVI) with temperature (more plants, cooler)

Across 111 wards

18.9 km²

is both 45 °C or hotter and low in vegetation (15% of the city)

Spatial overlap of two rasters

23 wards

are in the highest stress class, covering only 8.1 km² (6.6% of the city)

Top fifth of UCSI scores (111 ÷ 5 = 22.2, so the top class has 23)

How to read: a correlation (r) close to +1 or −1 means a strong relationship; close to 0 means almost none.

Meet the data

All of it is free

Landsat 8 and 9

A thermal camera in space.

Heat · 30 m (thermal 100 m)

Used: 13, 22 and 30 Apr 2026

FREE

Sentinel-2

Sharp colour + infrared photos.

Vegetation · 10 m

Used: 8 and 11 Apr 2026

FREE

ESA WorldCover

A map of what covers the ground.

Built-up, trees, water · 10 m

WorldCover 2021 (v200)

FREE

SRTM elevation

The height of the ground.

Elevation, slope · 30 m

NASA / USGS, year 2000

FREE

OpenStreetMap

A free map made by volunteers.

Major roads · vector

Credit: © OpenStreetMap contributors

FREE

Ward boundaries

111 ward polygons.

OpenCity / BharatLAS

Already in your GeoPackage

FREE

Earth Engine hosts Landsat, Sentinel-2, WorldCover and SRTM. The five scripts in the project folder (gee/) do the same steps but have not been run yet. The rasters in the project came from the same datasets on other public servers.

Indicators 1 and 2: heat and green cover

Heat: LST

Land Surface Temperature

THINK OF IT AS

A thermal camera photo of the ground.

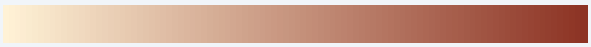
HOW WE GET IT

Landsat thermal band converted to °C. We take the median of 3 clear April dates so one odd day does not decide the answer.

REAL RESULT

33.7 to 47.6 °C

ward averages; the whole city averages 41.0 °C.



34 °C 48 °C

Careful: this is the temperature of the surface (roofs, roads, soil), not the air.

Green cover: NDVI

Normalised Difference Vegetation Index

THINK OF IT AS

A plant health meter. Leaves reflect lots of near-infrared light; roofs and roads do not.

FORMULA

`(NIR − Red) ÷ (NIR + Red)`

REAL RESULT

0.12 to 0.49

ward averages; 40% of the area is below 0.2.



below 0.2 bare or built-up

0.2 to 0.4 sparse plants

above 0.4 dense plants

April is dry, so some plants look brown. The result is one season only; a second season is a next step.

Indicators 3 and 4: built-up land and flood

Built-up land

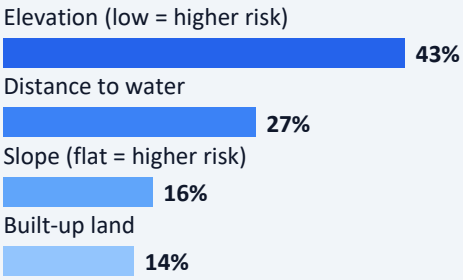
The share of each ward covered by buildings and pavement (WorldCover class 'Built-up').

33% of the study area is built-up

30 of the 111 wards are 80% or more built-up.

Flood susceptibility

How easily water can collect. It is a susceptibility score, not a flood forecast.



Weights from AHP: consistency ratio 0.017 (below 0.10 is acceptable).

Why flooding matters here: 26 May 2025

In about 8 hours (8:30 am to 4:30 pm) Nerul received 155.7 mm of rain and Belapur 133.9 mm, against an average of 80.8 mm across six nodes. Thane-Belapur Road, Sion-Panvel Highway, MIDC roads and the APMC market were waterlogged.

Source: Free Press Journal, 26 May 2025 (news report)

Why not rainfall as a factor?

Free rainfall data (CHIRPS) has 5-km pixels. Only about 4 of them cover the whole city, so every ward would get almost the same value. It cannot separate wards, so we left it out.

How the score is built

Example: ward 70, the highest-stress ward

1

Stretch each indicator to a 1–5 scale

Coolest ward = 1, hottest ward = 5. Same for vegetation, upside down: least plants = 5.

2

Score flood from four factors

Elevation, distance to water, slope and built-up land are scored 1–5, then weighted (43 / 27 / 16 / 14).

3

Average the three scores

UCSI = (Heat + Green-cover deficit + Flood) ÷ 3

WARD 70

Heat	Green-cover deficit	Flood
47.6 °C	NDVI 0.12	4 factors
5.00	**5.00**	**3.45**

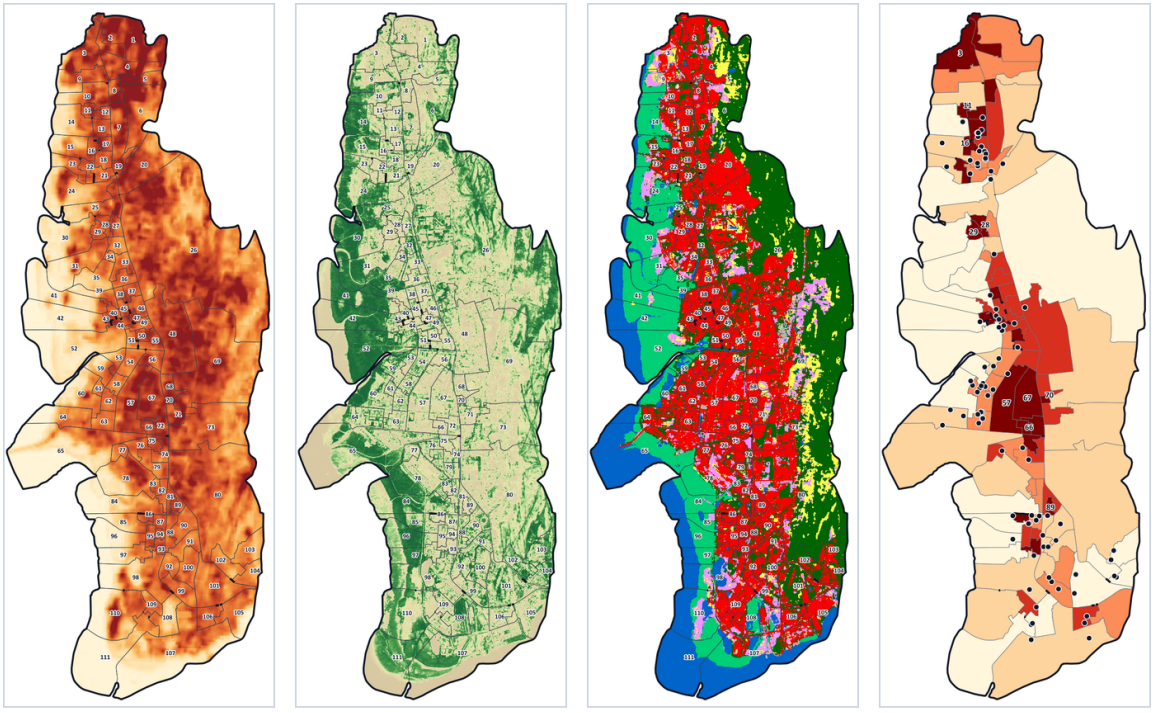
UCSI = (5.00 + 5.00 + 3.45) ÷ 3 = 4.48

Does the ranking change if the weights change?

We tried heat-heavy weights (50 / 25 / 25) and flood-heavy weights (25 / 25 / 50). The ward ranking hardly changes: rank correlation 0.983 and 0.95 (1.0 = identical), and 7 and 8 of the top 10 wards stay in the top 10.

Results: the four maps

Navi Mumbai, April 2026 (exported from the QGIS project)



A Heat
darker = hotter ground

B Green cover
darker green = more plants

C Land cover
red = built-up, green = trees

D Stress index
dark red = highest stress

A: darker = hotter surface. B: darker green = more plants. C: land cover (red = built-up, green = trees, blue = water). D: darker red = higher combined stress. Each map is a real satellite result, not a sketch.

Heat map: the north and centre are hottest

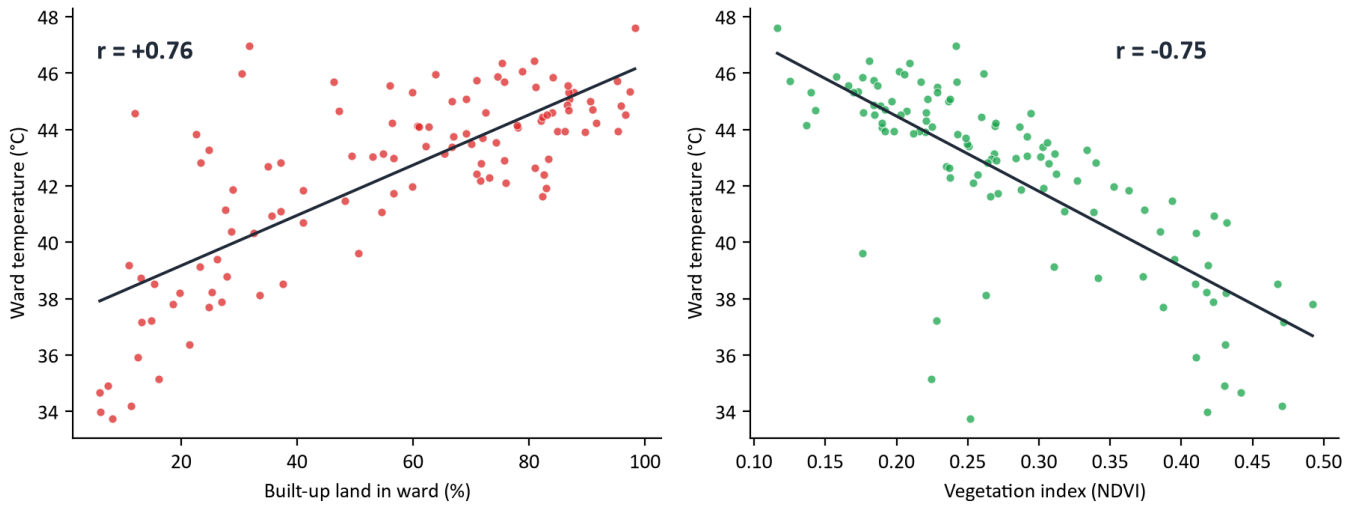
Green map: mangroves and hills stay greener

Stress map: 23 wards are in the highest class

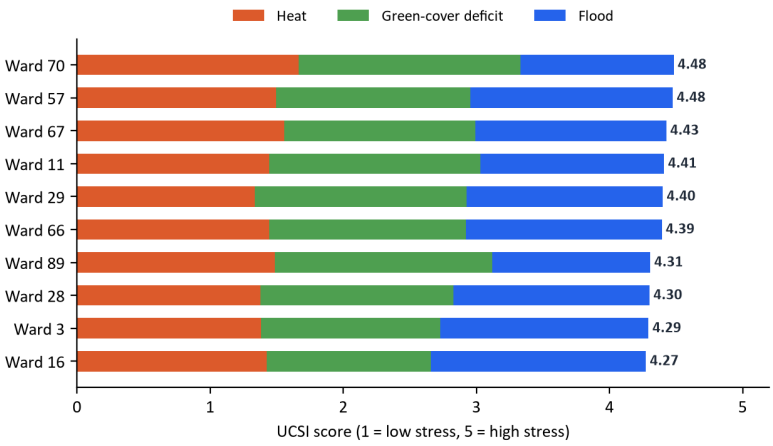
Watch out (limits)

In April the dry hills in the east also look hot, even though they are not built-up. The stress index (D) stays low there because the flood score is low on high ground. A second season is a next step.

Results: what the numbers say



Each dot is one ward. The line shows the trend.



Reading the top 10

Top wards: 70, 57, 67, 11, 29.
Ward 70 is tiny (0.07 km²), so treat it with care.
Wards 57 and 66 are in the top quarter for heat, green-cover deficit and flood at the same time.

What this means in plain words

- More built-up land in a ward goes together with a hotter ward (r = +0.76).
- More plants goes together with a cooler ward (r = −0.75).
- Correlation shows a link, not proof of cause, but it matches what we expect from heat-island studies.
- A next step is to repeat this in another season.

Ten queries and their answers

Run in QGIS (thresholds chosen after looking at the data)

5 ATTRIBUTE QUERIES (search by data, Select by Expression)

1 Built-up 80% or more — **30 wards**

`"BuiltUp_Pct" >= 80 wards 11, 12, 17, 28, 37, 38, 40, 43, 44, 45, 46, 47, 50, 57 ...`

2 Vegetation (NDVI) below 0.2 — **24 wards**

`"NDVI" < 0.2 wards 2, 8, 11, 22, 28, 29, 38, 40, 43, 44, 47, 50, 57, 66 ...`

3 Temperature 45 °C or more — **23 wards**

`"LST_C" >= 45 wards 1, 2, 4, 5, 7, 8, 11, 12, 16, 17, 19, 44, 48, 50 ...`

4 Flood score 4.5 or more — **22 wards**

`"Flood_Score" >= 4.5 wards 3, 9, 13, 15, 16, 23, 33, 34, 36, 39, 53, 56, 57, 60 ...`

5 Built-up 80%+ and NDVI below 0.2 — **18 wards**

`"BuiltUp_Pct" >= 80 AND "NDVI" < 0.2 wards 11, 28, 38, 40, 43, 44, 47, 50, 57, 66, 67, 70, 75, 81 ...`

5 SPATIAL QUERIES (search by location: Select by Location, Buffer)

1 Wards touching water or mangrove (1 ha or more) — **48 wards**

Select by Location: wards intersect water_bodies (Area_ha >= 1)

2 Wards within 200 m of a major road — **78 wards**

Buffer 200 m around motorway / trunk / primary roads, then Select by Location

3 Hospitals in the highest-stress wards — **8 of 93**

Select by Location: hospitals are within the 23 wards of the highest class

4 Nearest hospital to those wards (mean) — **693 m**

Centroids + Join attributes by nearest; longest 2505 m (ward 2), straight line

5 Ward with the waterlogged APMC site — **ward 57**

Select by Location: wards contain the flood_reports point (this ward is UCSI rank 2)

GeoPackage layers

Wards (111) · Water bodies (100) · Major roads (729) · Hospitals (93) · Flood reports (3) · Hot and low-green zones (71) · LST by date (table) · 9 rasters. Extra analysis: the raster overlay of hot (45 °C+) and low-green (NDVI < 0.2) land is 18.9 km², 15% of the city.

Presenting in 5 minutes

Slide by slide: what to say in one line

1 Title 0:15	Name, topic, tools (Earth Engine and QGIS).
2 Problem 0:45	Heat gap of 13.9 °C, 26% tree cover, Nerul 155.7 mm rain. Wards are not equal.
3 Aim and objectives 0:35	One index (UCSI) for 111 wards. Four objectives.
4 Data and method 1:00	Free data, scores 1 to 5, UCSI = average of heat, green deficit, flood.
5 Results 0:55	Four maps. Built-up +0.76 with temperature, vegetation −0.75. Say: one season, ward level.
6 Database and queries 0:50	One GeoPackage. 30 wards are 80%+ built-up; 78 wards are near major roads.
7 Output and conclusion 0:40	What is built, next steps. A planning aid, not a prediction.

Tips

- Say the numbers slowly and pause after 13.9 °C and 155.7 mm.
- Do not read the slides. Look at ma'am and glance at the map.
- If you do not know: “Good question. My current approach is X, and I will check it next.”
- Stop after the last line and let ma'am speak first.

Questions a teacher may ask

Short, simple answers

Why Navi Mumbai?
A planned city with 111 wards, real heat and flood problems, and free data for every layer.

What is LST? Is it air temperature?
Land Surface Temperature: how hot the ground surface is, seen by a satellite. Not air temperature.

Why April?
It is dry and clear before the monsoon, so satellite images are clean. One season only; a second season is a next step.

What is NDVI?
A plant index from red and infrared light. Low means bare or built-up; high means dense plants.

Why is rainfall not used?
Rainfall data has 5-km pixels. About 4 cover the city, so all wards look the same.

How did you choose weights?
Flood weights come from AHP (consistency ratio 0.017). UCSI uses equal weights, and we tested other weights: the ranking barely changes.

Is UCSI a prediction?
No. It compares wards using satellite indicators. It is a planning aid.

Where is Earth Engine used?
Five scripts in the folder gee/ collect Landsat, Sentinel-2, WorldCover and SRTM. They are written but not run yet. The rasters came from the same data, processed with Python. Say this honestly.

Did you draw the layers by hand?
No. The ward polygons and the OpenStreetMap layers were imported. Only the 3 flood-report points were digitised, from a news report.

Why wards and not the 8 zones?
Your data has ward numbers only. Add NMMC's ward-to-office list and the results can be grouped into 8 offices.

How reliable is a tiny ward?
Less reliable. Ward 70 covers 0.07 km² (about 76 heat pixels). Say so if it is asked.

What is next?
Run the Earth Engine scripts and compare the numbers, add a second season, and add the ward-office list.

Limits and next steps

What a careful teacher could notice, and what to do

Limits (say them first)

- One season (April 2026). Dry hills also look hot.
- Heat data is 100 m resampled to 30 m; tiny wards are less reliable.
- Surface temperature is not air temperature.
- The 8 ward offices are not in the data yet.
- Flood factors use SRTM (includes buildings) and no tide or drainage data.
- Area figures differ by source: your 111 ward polygons add up to 122.9 km².
- The boundary file is not an official NMMC dataset (your own note).

Next steps (do these)

- Register a Google Earth Engine project (non-commercial, free, no billing) and upload the wards as an asset.
- Run the five scripts in the folder gee/ and compare with the delivered values (mean LST about 41 °C, NDVI about 0.30). Then update the report's Earth Engine note.
- Open NaviMumbai_UCSI.qgz and run 2 or 3 queries by hand, so you can show them live.
- Add a second season (for example Nov–Feb) and compare.
- Copy NMMC's ward-to-office list into a table and group the results by office.
- Read the practical report once and mark anything you cannot explain.

Status: what is done and what is left

DONE Heat, green cover, built-up, flood score, UCSI, GeoPackage, 10 queries, 10 maps, practical report.
NEXT Run the Earth Engine scripts; try queries by hand in QGIS.
NICE Second season, 8 ward-office grouping.

Jargon in plain words

LST
Land Surface Temperature: how hot the ground surface is.

NDVI
Plant index. Higher = more plants.

UCSI
Urban Climate Stress Index: average of heat, green deficit and flood scores.

Ward
A small civic area. Navi Mumbai has 111.

Ward office
One of 8 administrative offices that group wards.

GeoPackage
One file that holds all map layers and tables.

Raster
A picture made of pixels with values (like heat).

Vector
Points, lines and polygons (like wards and roads).

Zonal statistics
Average of a raster inside each polygon (mean heat per ward).

Weighted overlay
Adding scores with different importance.

AHP
A method to decide weights by comparing factors two at a time.

Consistency ratio
Check that the AHP comparisons agree. Below 0.10 is fine.

Sensitivity check
Change the weights and see if the ranking changes.

Correlation (r)
−1 to +1. Shows how two things move together.

Attribute query
Search the table (for example built-up ≥ 80%).

Spatial query
Search by location (within, near, overlapping).

Buffer
A zone around a feature, like 200 m around a road.

CRS (EPSG:32643)
The map ruler. UTM zone 43N measures in metres.

Landsat / Sentinel-2
Free satellites: Landsat sees heat; Sentinel-2 sees detail.

Earth Engine
Google's free cloud tool for satellite data (non-commercial).

Appendix: numbers for all 111 wards

LST in °C; NDVI; built-up in %; flood and UCSI on the 1–5 scale

Ward	LST	NDVI	Bu%	Flood	UCSI	Ward	LST	NDVI	Bu%	Flood	UCSI	Ward	LST	NDVI	Bu%	Flood	UCSI
1	47.0	0.24	32	2.10	**3.53**	38	43.9	0.20	95	4.41	**4.16**	75	45.6	0.19	87	4.14	**4.27**
2	45.6	0.17	56	3.58	**4.15**	39	38.5	0.41	38	4.55	**2.94**	76	43.4	0.30	67	4.57	**3.79**
3	44.7	0.21	47	4.69	**4.29**	40	43.9	0.19	85	4.14	**4.09**	77	42.3	0.24	73	4.84	**4.00**
4	45.7	0.24	46	2.83	**3.65**	41	34.7	0.44	6	4.26	**2.35**	78	40.4	0.39	29	4.55	**3.20**
5	46.0	0.26	30	2.10	**3.36**	42	34.0	0.42	6	4.26	**2.37**	79	43.0	0.28	57	4.69	**3.86**
6	44.6	0.29	12	1.95	**3.06**	43	44.8	0.19	96	4.14	**4.19**	80	42.8	0.26	23	2.26	**3.10**
7	45.7	0.22	76	3.72	**4.03**	44	45.4	0.17	97	3.87	**4.21**	81	44.7	0.19	91	3.72	**4.03**
8	45.7	0.18	71	3.99	**4.24**	45	43.9	0.22	90	4.14	**3.99**	82	43.0	0.30	53	4.27	**3.66**
9	41.9	0.29	29	4.55	**3.69**	46	43.9	0.22	86	4.14	**4.01**	83	42.9	0.27	76	4.41	**3.81**
10	41.8	0.36	41	4.27	**3.33**	47	44.9	0.18	87	3.87	**4.12**	84	35.9	0.41	13	4.41	**2.63**
11	45.3	0.14	88	4.14	**4.41**	48	46.1	0.20	79	3.56	**4.07**	85	37.7	0.39	25	4.55	**2.94**
12	45.1	0.22	87	4.41	**4.19**	49	43.7	0.25	72	4.14	**3.87**	86	44.2	0.19	92	4.14	**4.13**
13	45.0	0.24	67	4.57	**4.18**	50	45.0	0.20	91	4.14	**4.18**	87	44.4	0.26	82	3.72	**3.76**
14	37.8	0.49	19	4.41	**2.53**	51	42.1	0.25	76	4.41	**3.79**	88	43.5	0.31	74	3.72	**3.51**
15	39.4	0.39	26	4.55	**3.07**	52	34.9	0.43	7	4.26	**2.42**	89	45.7	0.13	95	3.56	**4.31**
16	45.1	0.24	69	4.84	**4.27**	53	41.1	0.34	55	4.69	**3.48**	90	42.8	0.31	72	3.56	**3.38**
17	45.5	0.23	81	4.14	**4.11**	54	42.2	0.33	72	4.41	**3.54**	91	42.0	0.35	60	3.41	**3.09**
18	43.8	0.29	67	3.98	**3.67**	55	44.6	0.22	73	4.25	**4.09**	92	42.4	0.31	71	3.56	**3.33**
19	45.3	0.23	60	3.10	**3.75**	56	44.2	0.27	56	4.69	**4.03**	93	44.7	0.14	87	3.56	**4.14**
20	44.1	0.27	61	2.70	**3.36**	57	45.8	0.18	84	4.57	**4.48**	94	44.5	0.20	97	3.99	**4.06**
21	43.4	0.25	62	3.98	**3.78**	58	41.9	0.30	83	4.41	**3.60**	95	44.5	0.18	83	3.72	**4.03**
22	44.1	0.19	78	4.41	**4.20**	59	38.8	0.37	28	4.39	**3.04**	96	34.2	0.47	11	4.41	**2.25**
23	41.1	0.32	37	4.55	**3.51**	60	37.9	0.42	27	4.55	**2.83**	97	36.4	0.43	21	4.55	**2.65**
24	39.2	0.42	11	4.41	**2.92**	61	41.6	0.27	82	4.14	**3.61**	98	39.1	0.31	23	4.39	**3.29**
25	41.2	0.37	28	4.13	**3.17**	62	43.0	0.27	83	4.14	**3.74**	99	43.1	0.31	55	4.69	**3.78**
26	43.3	0.33	25	1.99	**2.81**	63	42.4	0.26	83	4.14	**3.72**	100	43.1	0.29	49	4.11	**3.64**
27	44.1	0.29	61	4.41	**3.86**	64	38.1	0.26	34	4.55	**3.42**	101	42.8	0.34	37	2.68	**2.97**
28	44.6	0.18	84	4.41	**4.30**	65	35.1	0.22	16	4.41	**3.22**	102	40.9	0.42	36	3.00	**2.60**
29	44.2	0.14	78	4.41	**4.40**	66	45.3	0.17	87	4.41	**4.39**	103	40.3	0.41	32	2.10	**2.29**
30	37.2	0.47	13	4.41	**2.54**	67	46.4	0.18	81	4.30	**4.43**	104	40.7	0.43	41	2.56	**2.40**
31	38.2	0.43	20	4.41	**2.78**	68	45.9	0.16	75	3.29	**4.12**	105	39.6	0.18	51	4.11	**3.72**
32	41.5	0.39	48	4.27	**3.18**	69	43.8	0.24	23	2.26	**3.27**	106	43.8	0.21	69	4.41	**4.11**
33	43.2	0.27	65	4.84	**3.98**	70	47.6	0.12	98	3.45	**4.48**	107	37.2	0.23	15	4.41	**3.41**
34	41.7	0.27	57	4.53	**3.73**	71	46.0	0.21	64	3.29	**3.95**	108	38.2	0.42	25	4.39	**2.82**
35	38.5	0.47	15	4.41	**2.68**	72	46.3	0.21	75	3.99	**4.21**	109	42.6	0.24	81	4.41	**3.90**
36	43.5	0.25	70	4.57	**3.99**	73	42.7	0.24	35	2.26	**3.19**	110	38.7	0.34	13	4.41	**3.15**
37	44.3	0.22	82	4.14	**4.03**	74	44.1	0.23	63	4.14	**3.99**	111	33.7	0.25	8	4.42	**2.99**

Red UCSI = the 23 wards of the highest class (rank 1 to 23). Values are from the QGIS project (April 2026). Ward numbers follow the GeoPackage (Ward_ID).

Sources and data notes

- Free Press Journal, 26 May 2025: Navi Mumbai rains, Nerul and Belapur worst affected (news report).
- Wikipedia, Navi Mumbai Municipal Corporation: 111 electoral wards (2026 elections).
- Landsat 8/9 Collection 2 Level 2 (USGS/NASA) via Microsoft Planetary Computer; scenes 13, 22 and 30 Apr 2026. Surface temperature = DN × 0.00341802 + 149.0 K.
- Sentinel-2 L2A (ESA/Copernicus) via Earth Search on AWS; scenes 8 and 11 Apr 2026.
- ESA WorldCover 10 m 2021 v200 (CC BY 4.0). Dynamic World (Google/WRI) is the Earth Engine alternative.
- SRTM 30 m (NASA/USGS). OpenStreetMap (ODbL, © OpenStreetMap contributors). Ward boundaries: OpenCity / BharatLAS files in your GeoPackage.
- Earth Engine noncommercial Community tier: 150 compute-hours a month, no billing account needed (Google documentation).
- Similar study (title consulted, not read in full): 'Identification of urban hotspot areas from remote sensing towards implementing heat mitigation strategies: a case study over Navi Mumbai', Proc. Indian National Science Academy, 2025.

Honest notes

• All numbers in this guide were calculated from the data above, not invented.
• The results are one season (April 2026) and ward level: a first screening, not a final assessment.
• The Google Earth Engine scripts are written but not run yet. The rasters came from the same datasets processed with Python.
• NMMC's own pages could not be read online (they need JavaScript), so the ward-office list is still to be copied by hand.
• Different sources give different NMMC areas (for example 108.6, 122.9 and 162.5 km²). We quote 122.9 km², the sum of your ward polygons.